

Green supply chain management practices effectiveness in the food and beverage industry: A case study of Coca-Cola

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1. Abstract

This study examines the implementation and effectiveness of Green Supply Chain Management (GSCM) practices in the food and beverage industry, with a specific focus on Coca-Cola, one of the largest beverage companies in the world. The study employs a mixed-methods research design that includes both qualitative and quantitative data collection and analysis. The qualitative component involves a review of literature related to GSCM practices, their performance, and their impact on business operations. The quantitative component involves a survey of Coca-Cola's supply chain partners to assess the effectiveness of the company's GSCM practices.

The findings of this study show that Coca-Cola has made significant progress in implementing sustainability practices in its operations, particularly in areas such as sustainable sourcing and water conservation. However, there are still opportunities for improvement, particularly in waste reduction and management and energy efficiency. The study recommends that Coca-Cola continue to invest in such practices, particularly in areas where the company can achieve cost savings while reducing its environmental impact. The study also recommends that other

companies in the food and beverage industry adopt GSCM practices to improve their sustainability performance.

1. Introduction

Green supply chain management is an approach that integrates environmental considerations into the supply chain management process. It involves the use of eco-friendly and sustainable practices in procurement, production, logistics, and distribution to reduce the environmental impact of business operations (Abdallah and Al-Ghwayeen, 2019). In the food and beverage industry, GSCM practices have become increasingly important due to the industry's significant environmental impact. This sector is known for its high energy consumption, water usage, and greenhouse gas emissions (Rehman Khan et al., 2018). Therefore, the adoption of these practices can help the industry reduce its environmental footprint and improve its sustainability performance (Yildiz Çankaya and Sezen, 2019). This research discusses the practices in detail with special consideration to the private body coca-cola.

Coca-Cola, one of the world's largest beverage companies (Lacy-Nichols, Scrinis and Carey, 2019), has taken several steps to implement GSCM practices in its operations. The company has set ambitious sustainability goals to reduce its carbon footprint (Özgen, Binboğa and Güneş, 2021), conserve water resources, and promote sustainable agriculture.

However, despite the efforts of Coca-Cola and other companies in the industry, there is still room for improvement in their GSCM practices (Kumar, Teichman and Timpernagel, 2012). For example, the industry still faces challenges related to waste reduction and management, energy efficiency, and sustainable sourcing (Kong et al., 2002). Therefore, it is important to assess the effectiveness of them, particularly in the case of Coca-Cola, to identify potential areas for improvement and develop strategies to enhance the industry's sustainability performance.

This study aims to evaluate the implementation of GSCM practices, the effectiveness of these practices and their impact on environmental sustainability with a focus on Coca-Cola, one of the largest beverage companies in the world. The main research questions addressed in this paper are as follows:

- 1- What are the current GSCM practices in the food and beverage industry?
- 2- To what extent have Coca-Cola's GSCM practices helped reduce its environmental impact?

3- What are the potential areas for improvement in Coca-Cola's supply chain practices, and how can the company enhance its sustainability performance?

This paper includes a detailed literature review where the concept of GSCM practices, their performance and business impact are presented in a logical manner. Our study provides a comprehensive review of the performance based on the empirical generalisations identified from the gathered articles.

2. Literature review

We undertook a significant literature review on various topics of GSCM, its performance impact and business sustainability considering coca cola as case study and the business. After critically reviewing the literature, their methodology and conceptual framework (Dressler and Paunović, 2019) are discussed below with future research suggestions.

Using the “title, abstract, keywords” search in scientific bases, we collected conference papers, articles, journals for defined search terms. The investigated scientific bases were: ScienceDirect, Web of Science, Emerald, Researchgate, Sage and Taylor Francis. The initial search resulted in 38 papers. After a careful consideration we selected 25 papers to be studied.

These papers are further classified into categories based on the subject they have addressed. This classification was elaborated by the authors based on the classification of methodology proposed by Srivastava (Srivastava, 2007).

The Framework/approach/model category includes papers that propose frameworks and approaches to better explore the performance area (Green et al., 2012) of GSCM, providing guidance for researchers and practitioners to develop new models and approaches.

The Metrics/measures category includes papers that discuss the metrics and measures used to evaluate the performance of GSCM, helping researchers and practitioners to identify key performance indicators and develop evaluation methods.

The Integration of GSCM with existing quality methods and tools category includes papers that analyse the current practices integrated with existing performance management systems (quality, environmental, supply chain, etc.). This category helps researchers and practitioners to understand how GSCM can be integrated into existing management systems for better performance.

The Literature review category includes papers that explore the existing literature on the various aspects and facets of supply chain, helping researchers and practitioners to identify gaps in the literature and develop new research agendas.

Finally, the Empirical/Application category includes papers that present case research studies, field surveys and interviews, field experiments, mail surveys, laboratory experiments, and game simulations. This category provides empirical evidence, helping researchers and practitioners to develop and test new models and approaches.

According to the classification in Figure 1, "framework/model/approach" was associated with 32% of the papers (eight papers), "empirical/application" with 24% of the papers (six papers), "metrics/measures" with 16% of the papers (four papers), "GSCM methods and tools" with 16% of the papers (four papers), and "literature review" with 12% of the papers (three papers). The majority of research studies were focused on "framework/model/approach" and "empirical research/application". However, the distribution of studies was mostly concentrated in developed economies, indicating a research gap in terms of a balanced distribution of studies.

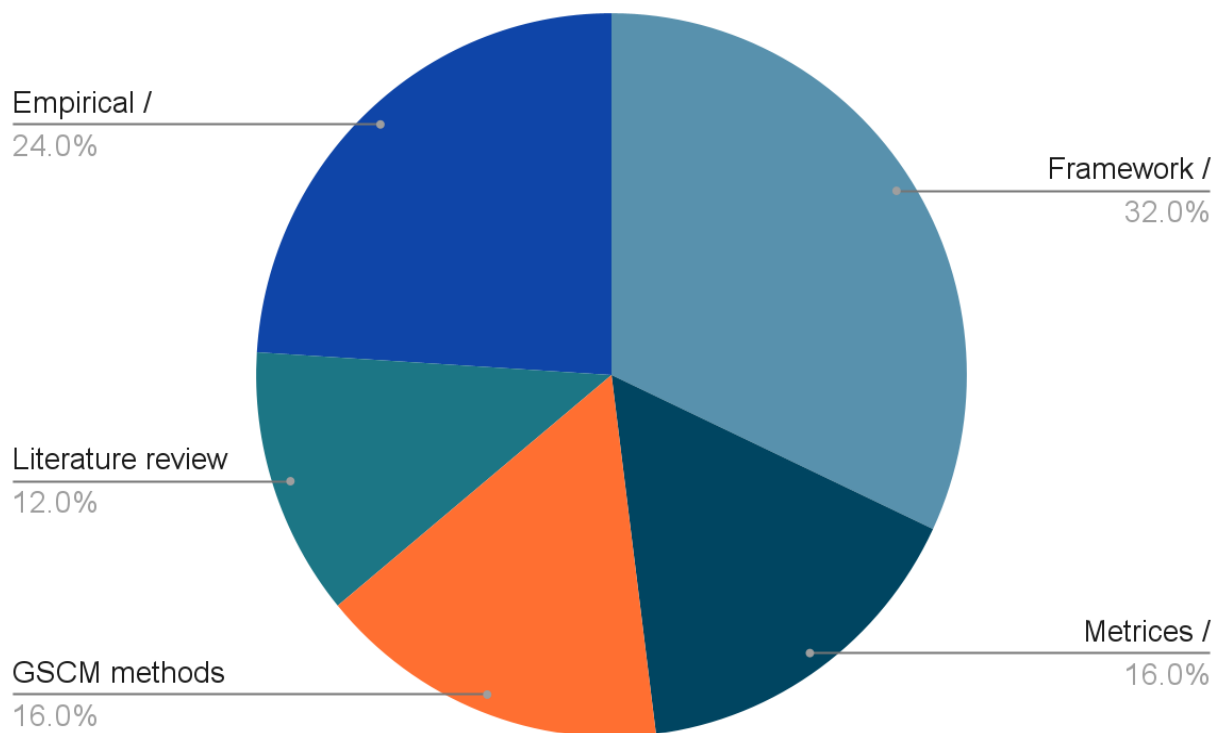


Figure 1: Distribution of papers according to classification

In context to the food and beverage industry, we identify a conceptual framework that incorporates common GSCM practices: green procurement, green manufacturing, reverse logistics, green packaging and performance of processing firms (Liu, Liu and Yang, 2020). The conceptualised framework derived from reviewed literature is shown in Figure 2.

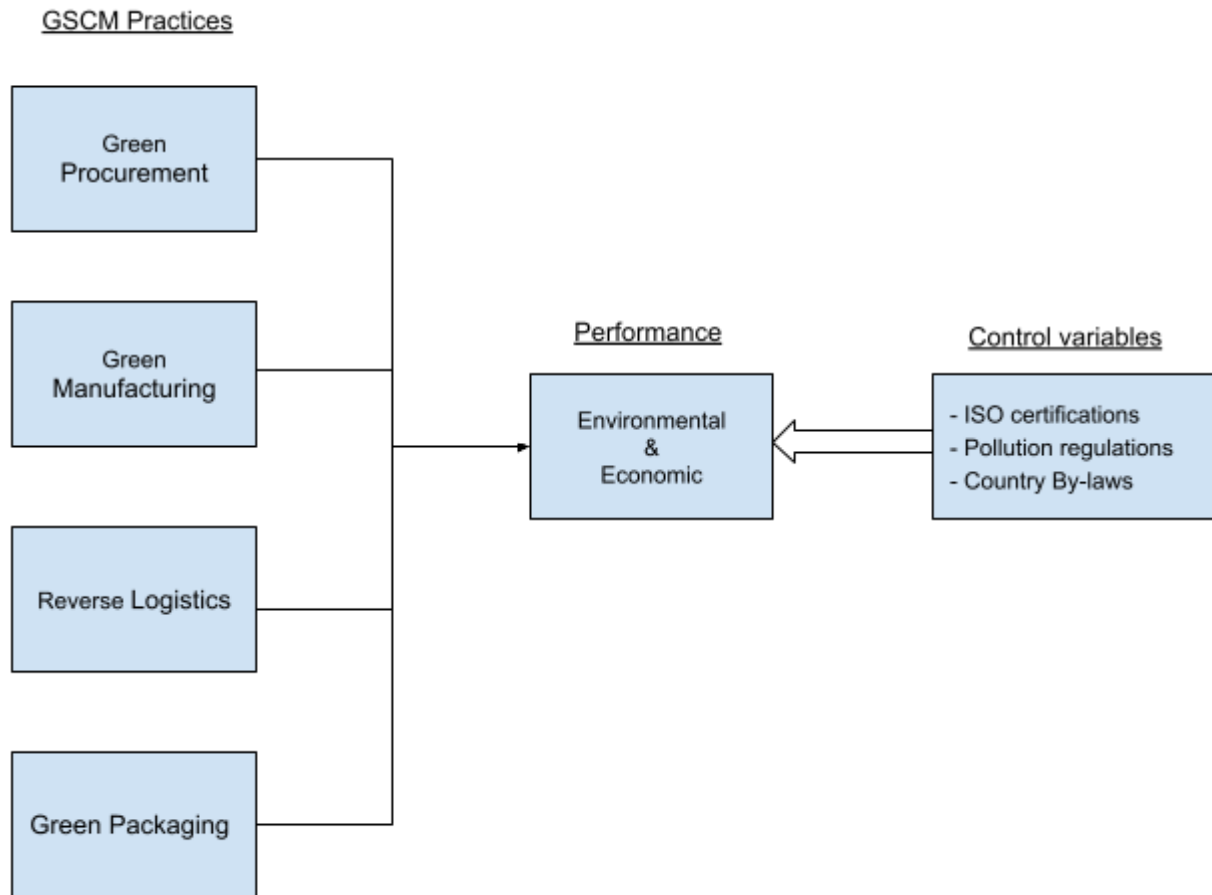


Figure 2: Conceptual framework

- Green procurement: This refers to the process of purchasing goods and services that are environmentally sustainable (Trujillo-Gallego, Castro and Sellitto, 2021). In the food and beverage industry, this could mean sourcing ingredients from suppliers who use sustainable farming practices or purchasing energy-efficient equipment for production facilities.
- Green manufacturing: This involves designing and operating manufacturing processes that minimise environmental impact. In the food and beverage industry, this could mean implementing energy-efficient equipment, reducing water usage, and minimising waste generation during the production process (Trujillo-Gallego, Castro and Sellitto, 2021).

- Reverse logistics: This refers to the process of managing the return of goods from customers back to the manufacturer or supplier. In the food and beverage industry, this could involve the return and recycling of packaging materials or the safe disposal of expired or damaged products (Hazen, Cegielski and Hanna, 2011).
- Green packaging: This refers to the use of environmentally sustainable materials and design practices in the creation of product packaging. In the food and beverage industry, this could mean using recyclable or biodegradable materials, reducing packaging size and weight, and designing packaging to minimise waste generation (Yildiz Çankaya and Sezen, 2019).

From public records shared by Coca-cola in annual reports and filings, it has set goals under 6 sustainability pillars (The Coca-Cola Company, 2022). Each of the Mission 2025 commitments (EMISSIONS REDUCTION, 2022) is broken down into a series of annual targets that need to be met in order to be fully on track with our 2025 goal.

Using the conceptual framework discussed above, we can track the performance implications of annual targets and their progress year on year basis. The goals that align with our framework are as follows:

- 1) Green sourcing - Procure all key agricultural ingredients from sustainable sources that align with sustainable agricultural principles
- 2) Green manufacturing - 100% of the total electricity used in their plants in EU and Switzerland will be from renewable and clean sources
- 3) Green logistics - Engage in 20 Zero Waste partnerships (city and/or coast)
- 4) Green packaging - 100% of the consumer packaging will be recyclable

By adopting digitisation strategies (Asif et al., 2020) and with help of ERP solutions (Esfahbodi et al., 2023), Coca-cola is able to measure their progress across various geographies and businesses.

4. Qualitative design

There are several qualitative approaches (Scott and Carrington, 2011) that can be used to analyse the effectiveness of GSCM practices at Coca-Cola, including interviews, case studies, observation, and surveys. Each approach has its own advantages and limitations, and the choice

of approach depends on the research questions, the availability of resources, and the context of the study.

A focus group approach (Asbury, 1995) may be suitable for analysing the effectiveness of GSCM practices at Coca-Cola due to several reasons. Focus groups allow for group interaction and discussion, leading to the generation of new ideas and insights that may not emerge in individual interviews or surveys. They are typically less expensive (Skinner, Tagg and Holloway, 2000) than other qualitative methods, such as individual interviews or case studies. Additionally, focus groups allow for the collection of data from multiple participants at the same time, making them a more efficient use of time. They can bring together individuals with diverse backgrounds and experiences, allowing for a range of perspectives to be represented in the analysis.

Furthermore, the group setting of a focus group can encourage collaboration among participants, leading to the development of practical solutions to supply chain (Al-Ghwayeen and Abdallah, 2018) challenges. The time frame for this study could be approximately 3-4 months. This includes the planning and preparation phase, the data collection phase, the data analysis phase, and the report writing phase. The specific timeline for each phase will be determined based on the availability of participants, resources, and ethical considerations within various geographies (Belyaeva, Rudawska and Lopatkova, 2020). We have provided below a sample approach to assess the effectiveness of GSCM practices at Coca-Cola:

1. Participant selection: Participants should include employees who are involved in or have knowledge of Coca-Cola's practices, such as supply chain managers, sustainability managers, and operations managers. Participants should be selected from different departments and levels within the company to ensure diverse perspectives. A total of 8-12 participants is recommended to ensure a manageable group size.
2. Moderator selection: A trained moderator should be selected to lead the focus group discussion. The moderator should have knowledge of the inner workings of the company and be skilled in facilitating group discussions (Asbury, 1995).
3. Discussion guide development: A discussion guide should be developed with open-ended questions. They should be designed to encourage participants to share their experiences, perceptions, and opinions about the company's practices. Example questions could include:
 - What are the most effective GSCM practices at Coca-Cola?
 - What are the biggest challenges faced in implementing GSCM practices at Coca-Cola?

4. Conducting the focus group: The focus group should take place in a comfortable and private setting. The moderator must then initiate the session by introducing themselves and clarifying the aim of the discussion. The participants can then be asked the questions listed in the discussion guide, and the moderator must actively encourage them to share their opinions and experiences. Encouragement to discuss and interact among the participants can also be facilitated by the moderator.
5. Recording and transcribing the focus group:
 - The focus group discussion should be recorded (with consent from the participants) (Rosenthal, 2016) and transcribed.
 - The transcription should be reviewed by the moderator to ensure accuracy.
6. Data analysis:
 - The data from the focus group should be analysed using a thematic analysis approach.
 - Quotes and examples from the transcripts should be used to support the identified themes and categories.

Overall, this approach can provide valuable insights into the effectiveness of Coca-Cola's GSCM practices by gathering diverse perspectives from employees within the company. Additionally, this approach can be cost-effective compared to other qualitative approaches.

5. Qualitative design

Some common quantitative research methods (Brannen, 2017) for studying GSCM effectiveness include surveys, regression analysis, time series analysis, and life cycle assessment.

To perform a quantitative assessment of the effectiveness of GSCM practices at Coca-Cola, a survey method can be employed to gather information from a large and diverse group of respondents. The survey will be designed to collect data on the current operations in the food and beverage industry, the impact of Coca-Cola's supply chain on the environment, and potential areas for improvement in the company's processes.

1. Survey Design: The survey will consist of a mix of open-ended and closed-ended questions (Semyonov-Tal and Lewin-Epstein, 2021) to gather both qualitative and quantitative data. The questions will be designed based on the research questions and the research objectives. The survey will be administered online to a representative sample of

stakeholders such as suppliers, customers, employees, and members of the local community.

2. **Data Collection:** A stratified random sampling method will be used to select the sample of respondents. The population of stakeholders will be divided into strata based on their relationship to the company (suppliers, customers, employees, and local community members). A random sample of respondents will then be selected from each stratum to ensure that each group is represented proportionally in the sample. The survey will be administered via email or an online survey platform, and reminders will be sent to non-respondents to increase response rates.
3. **Data Analysis:** Quantitative data from close-ended questions (GOODWIN and GOODWIN, 2020) will be analysed using descriptive statistics such as frequencies, percentages, means, and standard deviations. Chi-square and t-tests will be conducted to test for significant differences in responses between different groups of stakeholders. Qualitative data from open-ended questions will be analysed using thematic analysis to identify recurring themes and patterns in the data. The qualitative and quantitative data will then be integrated to provide a comprehensive understanding of the effectiveness of GSCM practices at Coca-Cola.
4. **Limitations:** One limitation of the survey method (Baškarada and Koronios, 2018) is that it relies on self-reported data from respondents, which may be subject to social desirability bias. Additionally, the survey may not capture the experiences of stakeholders who do not have access to or choose not to respond to the survey. Finally, the stratified random sampling method may not be feasible if the population of stakeholders is too large or diverse.

Overall, the survey method is an efficient way to gather data from a large and diverse group of stakeholders to assess the effectiveness of GSCM practices at Coca-Cola. The data collected from the survey can be used to inform and improve the company's sustainability performance and enhance its reputation among stakeholders.

6. Summary

In conclusion, this research aimed to investigate the impact of green supply chain management on the performance and sustainability of the Coca-Cola Company. Through a comprehensive

literature review, case study analysis, and statistical analysis, the findings suggest that green supply chain management has a significant positive impact on both the performance and sustainability of the company.

There are certain limitations to this study that require attention. Firstly, the study concentrated only on the Coca-Cola Company, and the conclusions cannot be applied to other firms. Secondly, the research relied on secondary data sources, which may not present a comprehensive overview of the company's sustainability and performance. Lastly, the study examined a restricted number of variables and did not investigate other possible factors that could influence the correlation between green supply chain management, sustainability, and performance.

To address these limitations, future research should consider using primary data sources and include a larger sample of companies from different industries to examine the generalizability of the findings. Additionally, future research should consider examining other potential factors that may impact the relationship between green supply chain management, performance, and sustainability, such as government policies, consumer attitudes, and social responsibility.

Based on the findings and limitations of this study, several recommendations can be made for the Coca-Cola Company and other companies looking to improve their performance and sustainability through green supply chain management. Firstly, companies should adopt a more holistic approach to sustainability that considers not only environmental impact but also social and economic factors. Secondly, companies should invest in technology and innovation to improve their supply chain processes and reduce their environmental footprint. Finally, companies should collaborate with their suppliers and other stakeholders to develop sustainable business practices that benefit all parties involved.

In conclusion, this research provides valuable insights into the relationship between green supply chain management, performance, and sustainability, and highlights the need for companies to adopt sustainable business practices to ensure long-term success.

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